

Q2(a)

Tangential speed of acrobat B is  $v = r\omega$

[1]

$$= (2.0 + 1.1) \times 1.290 = 3.999 = \mathbf{4.00 \text{ m s}^{-1}}$$

[1]

Q2(b)(i)

The net vertical forces on A towards the centre of the circle provide the centripetal force for circular motion of acrobat A.

$$mg - N = \frac{mv^2}{r}$$

When acrobat A just loses contact with the cage, contact force  $N = 0$

$$mg - 0 = \frac{mv^2}{r} \quad [1]$$

$$g = \frac{v^2}{r}$$

$$9.81 = \frac{v^2}{2.9} \quad (\text{c.g. of A is 2.9 m from the centre}) \quad [1]$$

$$v = 5.333 = \mathbf{5.33 \text{ m s}^{-1}} \quad [1]$$

Q2(b)(ii)

Both A and B are moving at the same angular velocity.

Since  $v = r\omega$ , we have

$$\frac{v_A}{r_A} = \frac{v_B}{r_B} \text{ or } \frac{5.33}{2.9} = \frac{v_B}{3.1}$$

$$v_B = 5.6975 = \mathbf{5.70 \text{ m s}^{-1}} \quad [1]$$

Q2(c)

The net vertical forces on A provide the centripetal force for its circular motion.

$$mg + N = \frac{mv^2}{r}$$

When acrobat A is just in contact with the top of cage, contact force  $N = 0$

$$mg + 0 = \frac{mv^2}{r} \quad [1]$$

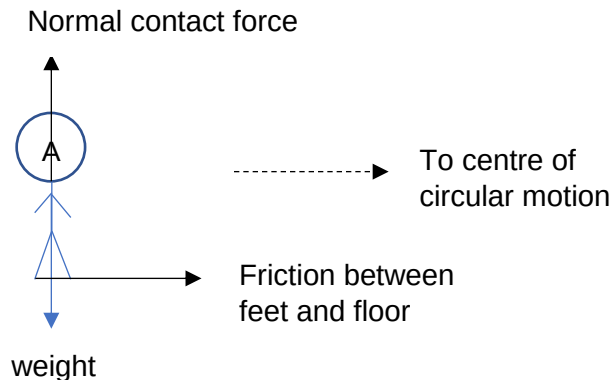
$$g = \frac{v^2}{r}$$

$$9.81 = \frac{v^2}{3.1} \quad (\text{c.g. of A is 3.1 m from the centre})$$

$$v = 5.5146 = 5.51 \text{ m s}^{-1}$$

Minimum velocity is **5.51 m s<sup>-1</sup>** [1]

Q2(d)



[2]

[ 1 mark for correct forces; 1 mark for identification of forces]